Project Plan

Contracts Information Management System (CIMS)

for the Philippine Center for Economic Development (PCED)

Client: Philippine Center for Economic Development
University of the Philippines School of Economics
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CS 191 WFQ1 - Group LF1
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1 Customer Statement of Requirements

1.1 Introduction

The Philippine Center for Economic Development (PCED) was established in 1974 and was given the key role of providing financial and moral support to the various programs of the UP School of Economics (Philippine Center for Economic Development, 2024).

Currently, PCED manages a variety of contracts and agreements that support its projects, partnerships, and organizational activities. These contracts include Memoranda of Agreement (MOAs), service contracts, partnership agreements, and other institutional commitments involving multiple offices and stakeholders. As the number of contracts managed by PCED continues to increase, the efficiency and reliability of handling these documents become even more crucial. Contract preparation, tracking, monitoring, and reporting require constant coordination among different offices within the organization. Ensuring that all stakeholders have access to accurate, complete, and up-to-date information is essential in maintaining efficient operations and supporting informed decision-making.

1.2 Problem

Currently, PCED's contract management process relies primarily on spreadsheets and other manual tracking methods. Although spreadsheets provide a convenient way to record information, they become increasingly difficult to maintain as the number of contracts grows and more stakeholders become involved in the process.

Since contract information is often stored across multiple files and managed by different offices and personnel, it can be challenging to ensure that everyone is working with the most recent version of the data. Frequent manual updates increase the risk of data inconsistencies, duplicate records, and missing information. As a result, tracking contract progress and maintaining accurate records can become time-consuming and prone to human error.

Additionally, the lack of a centralized platform that supports the entire contract lifecycle remains a significant challenge. Without a centralized system that minimizes data inconsistencies, duplicate records, and missing information, contract management can become inefficient and difficult to coordinate across multiple offices. Monitoring contract status and retrieving important information may require significant manual effort, reducing productivity and increasing the possibility of oversights.

1.3 Proposed Solution

To address these issues, the Contracts Information Management System (CIMS) will be developed to provide PCED with a centralized platform for preparing, tracking, and managing contracts and other formal agreements. One of the system's primary goals is to standardize the contract preparation and management process. CIMS will allow users to create and manage templates for commonly used contract types. These templates may include Memoranda of Agreement (MOAs), partnership agreements, and other recurring institutional documents. By allowing the easy creation and management of standardized contract templates, users can reduce the time spent preparing contracts while also ensuring consistency and efficiency across similar agreements.

Aside from contract preparation, CIMS will support the execution and management stages of the contract lifecycle. Users will be able to record and manage contract information, identify the stakeholders involved in each agreement, update contract records, and monitor the progress and status of contracts from creation to completion. Additionally, the system will support easier integration into this centralized platform by allowing the importation of spreadsheets and CSV files, which can then be converted into properly structured entries within the database.

The system will also support role-based access control to ensure that users can only access and modify information relevant to their responsibilities. Depending on a user's office, department, or position, permissions can be assigned to prevent unauthorized modifications and reduce the risk of accidental changes to contract records.

An initial implementation of CIMS has already been developed by previous student developers. However, the current system has yet to be fully deployed to the PCED server for organizational use. Furthermore, additional functionalities and feature revisions are still needed to better support the contract execution and management process.

Another major focus of this project will be the creation of comprehensive documentation. This includes both technical documentation for future developers and user documentation for PCED personnel. Proper documentation will make it easier to maintain and further enhance the system in the future, while also helping users effectively utilize the platform in their day-to-day operations. Through these improvements, CIMS aims to provide PCED with a reliable, maintainable, and user-friendly centralized contract management system that improves efficiency, promotes consistency, and supports the organization's contract management needs.

1.4 Project Priorities

Based on initial discussions with the client, the project's immediate priorities are:

1. Understand the existing implementation and evaluate its current capabilities.
2. Deploy and stabilize the current working prototype.
3. Implement additional contract execution and management features.
4. Improve user management and access control mechanisms.
5. Fix existing issues and improve system usability.
6. Produce comprehensive technical and user documentation.
7. Ensure that future developers and PCED personnel can easily maintain and use the system.

1.5 Stakeholders

The primary system owner will be the PCED Special Projects Office. However, the system is expected to be used by personnel from multiple offices within PCED. These users will participate in contract preparation, monitoring, approval, management, and reporting activities.

The stakeholders include:

- PCED Special Projects Office
- Administrative personnel
- Project coordinators
- Other PCED offices involved in contract-related workflows

1.6 Technical Context

Previous student developers developed the current CIMS implementation, which is available as a partially completed prototype. Initial findings suggest that the system utilizes SvelteKit for the web application framework and Supabase for database management and backend services.

1.7 Preliminary Requirements List

1.7.1 Functional Requirements

1. The system shall allow authorized users to create and manage contract records.
2. The system shall support the preparation of different contract and agreement types.
3. The system shall track the status of contracts throughout their lifecycle.
4. The system shall maintain a centralized repository of contract information and supporting documents.
5. The system shall support searching, filtering, and retrieval of contract records.
6. The system shall provide role-based access control.
7. The system shall allow administrators to manage user accounts and permissions.
8. The system shall maintain an audit trail of contract-related activities.

1.7.2 Non-Functional Requirements

1. The system shall be accessible through modern web browsers.
2. The system shall provide an intuitive and user-friendly interface.
3. The system shall maintain data consistency across all users.
4. The system shall securely store contract information and user data.
5. The system shall be maintainable and properly documented for future development teams.

1.8 Preliminary UI Design

Unfortunately, at the time of writing, a complete evaluation of the existing user interface could not be conducted due to issues with the system's backend infrastructure. The current implementation of CIMS relies on a Supabase backend; however, initial attempts to run the application revealed that the configured Supabase instance is currently inaccessible.

Testing showed that the application is unable to establish a connection with the backend services because the configured Supabase project endpoint cannot be reached. As a result, authentication and data retrieval functionalities are unavailable, preventing the development team from fully accessing and evaluating the current implementation.

Because of this limitation, detailed UI evaluation is currently pending. The team is coordinating with previous developers to obtain the necessary backend access, configuration details, and documentation required to restore the system to a working state.



Figure 1.8. Preliminary UI Design.

2 Plan of Work

2.1 Estimate Cost and Resources

2.1.1 Estimation Techniques

The two estimation techniques that will be used to estimate the project's size and effort are Function-Point Analysis and Process-Based Estimation (R. Pressman, 2009).

2.1.1.a Function-Point Analysis

Table 2.1.1.a. Function-Point Analysis.

Information Domain Value	**Opt.**	**Likely**	**Pess.**	**Est. Count**	**Weight**	**FP Count**
Number of External Inputs	6	8	10	8	4	32
Number of External Outputs	4	5	7	5	5	25
Number of External Inquiries	5	6	8	6	3	18
Number of Internal Logical Files	3	5	7	5	15	75
Number of External Interface Files	1	2	3	2	5	10
Count Total						**160**

The number of external inputs, external outputs, external inquiries, internal logical files, and external interface files expected from the system were estimated based on the initial requirements and existing implementations provided by Dr. Lagman. With the following estimations, the initial functional size of CIMS is therefore 160 function points (FPs). As the team’s scope is not to create the system from scratch and instead to add several features, then verify, document, and deploy the existing system to the PCED server, we will roughly estimate a value adjustment factor of 0.6 for the requirements entrusted to our team.

Multiplying our FP Count with the value adjustment factor, we get our FP of 96 effective FP. Since our team does not possess historical productivity data during the initial development of CIMS, we instead take an estimate that our team may accomplish 9 FP/Month (Azam, 2026). Therefore dividing 96 FP by 9 FP/Month, our team may be expected to accomplish the project in 10.67 months, or to an estimate of around 11 months.

2.1.1.b Process-Based Estimation

Table 2.1.1.b. Process-Based Estimation.

Development Activity	**Estimated Effort**
Existing-system evaluation and requirements clarification	25 person-hours
Deployment and stabilization of current working prototype	50 person-hours
Additional implementation of contract execution and management features	40 person-hours
UI improvements and system usability fixes.	35 person-hours
Testing and debugging	35 person-hours
Deployment and stabilization of applied and existing features.	35 person-hours
Technical and user documentation	30 person-hours
Total Estimated Effort	**250 person-hours**

From the given estimated effort per listed tasks, the total estimated effort to accomplish the project will be approximately 250 person-hours.

2.1.2. System Resources

Development will require computers capable of simultaneously running the web development environment, browser, source-code tools, and other supporting development software (R. Pressman, 2009). Thus, the deployed system will require:

- A server capable of hosting the CIMS web application
- Network connectivity between the application and its backend services
- Adequate storage for contract information and supporting documents
- A reliable network connection for users accessing the web application
- Modern web browsers on client computers

Exact server specifications will be determined after the team obtains the PCED deployment environment and evaluates the requirements of the existing implementation.

2.2 Identify Schedules and Milestones

The schedule below integrates the course deliverables from the CS 191 syllabus for the first semester with the projected development milestones extending into the CS 192 second semester.

Table 2.1a. First Semester: CS 191 (August 10 - December 5, 2026) (University of the Philippines Diliman OUR, 2026).

Target Date	**Deliverable/Milestone**
Sep 11, 2026	Project Plan
Sep 23, 2026	System Requirements Document (Part 1)
Sep 25, 2026	Existing system evaluated
Oct 9, 2026	System Requirements Document (Part 2)
Oct 16, 2026	Requirements & Software Requirements Specification finalized
Nov 6, 2026	System design complete
Nov 11, 2026	Initial Software Release (video demo)
Nov 25, 2026	Software Test Report
Nov 27, 2026	Prototype deployed & stabilized
Dec 5, 2026	Deliverables finalized

Table 2.1b.Second Semester: CS 192 (January 15 - May 19, 2027) (University of the Philippines Diliman OUR, 2026).

Target Date	**Deliverable/Milestone**
Feb 12, 2027	Completion of contract execution and management features
Mar 8, 2027	Completion of the user management and role-based access control modules
Apr 9, 2027	Completion of the reporting and dashboard modules
Apr 30, 2027	Finalized system testing, delivery of the test report, and client sign off
May 10, 2027	Finalization of documentation, the users' manual, and implementation guides
May 19, 2027	Presentation of the final system and handover of the full documentation set

2.3 Project Metrics

In order to measure the status of the project and avoid any possible delays, we will use the following metrics (R. Pressman, 2009):

1. Task Accomplishment Rate – checks how much of the tasks for the week the team has accomplished.

   This is computed by:

   $$Task\ Accomplishment\ Rate = \frac{Completed\ Tasks}{Total\ Tasks}$$

   where each development activity will have its needed tasks to be accomplished in order to track accomplishment rate.

2. Schedule Variance – checks whether the team is on track with the proposed project schedule.

   This is computed by mapping every task to a value, where:

   `-1` – the task was completed behind schedule
   `0` – the task was completed on schedule
   `1` – the task was completed ahead of schedule

   Schedule Variance will be a running total which will inform the team of project status accordingly.

2.4 Effort Breakdown

While all three members will jointly participate in general development, requirements gathering, system design, and coding across all modules, specific coordination responsibilities are divided as follows:

- Dominic Miguel D. Ceñido: Primary client contact and liaison.
- Bernadette T. Bernardino: Documentation coordinator, responsible for compiling technical and user documentation.
- Russell Lawrence N. Sison: Testing coordinator, responsible for system testing, bug tracking, and producing the Software Test Report.

3 Project Risks

Table 3. Project Risks (R. Pressman, 2009).

Risks		Category	Probability	Impact	RMMM
01	Difficulty understanding existing implementation	Technical	70%	2	**Mitigation**: Evaluate the existing system early and review existing project documentation. **Monitoring**: Track unresolved technical questions and implementation uncertainties. **Management**: Consult previous developers and the client when needing clarification.
02	Difficulty in deployment to server	Technical	60%	2	**Mitigation**: Begin deployment preparation early and document the server requirements. **Monitoring**: Track deployment and configuration. **Management**: Prioritize working towards developing the server deployment and consult when needed.
03	Inability to contact previous student developers	Stakeholder	30%	3	**Mitigation**: Contact student developers early for easier contact. **Monitoring**: Track technical questions that may arise from the existing implementation. **Management**: Rely on the available source code and client-provided information, and consult the internet.
04	Time constraints to meet the client's needs	Project/Schedule	50%	2	**Mitigation**: Prioritize deployment of the existing CIMS system to the server to identify early issues. **Monitoring**: Use the project metrics. **Management**: Prioritize implementing the core priorities for the project.
05	Inexperience in project implementation as specified by client	Stakeholder	30%	2	**Mitigation**: Allocate time for understanding the existing SvelteKit and Supabase implementation. **Monitoring**: Track unfamiliar syntax and other technical

					matters. **Management**: Ask the client for help.
06	Requirements change during development	Project	40%	2	**Mitigation**: Validate requirements with the client before major implementation work. **Monitoring**: Frequently meet with the client to ensure project alignment. **Management**: Prioritize changes according to their importance and available project time.

Where impact values are:

1 – catastrophic
2 – critical
3 – marginal
4 – negligible

4 References

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